

Protein and Nucleotide Foundation Models Encode Orthogonal Signals for Variant Pathogenicity

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Abstract

Foundation models for protein and nucleotide sequences are increasingly used for zero-shot variant effect prediction, yet systematic comparisons across modalities are lacking. We benchmark 11 protein and nucleotide foundation models on 29,595 ClinVar missense variants under a strict gene-disjoint evaluation protocol, and validate on a temporally independent dataset (9,627 variants, Jan 2025). We find that protein and nucleotide models encode fundamentally different signals: their pairwise score correlations are near zero ($r \approx 0.00$), confirmed by CKA (mean cross-modality CKA = 0.031). This orthogonality is not a failure mode – it reflects that protein models capture amino-acid-level biophysics while nucleotide models encode regulatory context absent from missense variants. We demonstrate this complementarity in two orthogonal directions. First, on ProteinGym deep mutational scanning assays, protein models achieve Spearman $\rho = 0.22$ – 0.32 while nucleotide models perform near chance ($\rho \leq 0.12$). Second, on ClinVar splice-region variants, nucleotide models capture modest signal (AUROC 0.640) while protein models are at chance (0.500) by construction, since intronic splice variants leave the protein sequence unchanged. We show that a lightweight meta-learner fusing protein scores with physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data. Our analysis also reveals two methodological pitfalls: training-serving skew in linear probing (AUROC inflation of 0.02–0.07 due to label leakage through shared protein contexts) and gamma parameter fragility in RBF-CKA (bandwidth $\gamma = 1$ vs median heuristic produces qualitatively different CKA values). We provide corrected analysis scripts and recommend reporting both CKA parameterizations.

1 Introduction

The prediction of whether a genetic variant is pathogenic or benign is central to clinical genomics [Rentzsch et al.(2019)Rentzsch et al.(2019)]. Recent foundation models trained on protein sequences [Elnaggar et al.(2022)Elnaggar, Heinzinger, Dallago, Rehawi, Wang, J. Lin et al.(2023a)Lin, Akin, Rao, Hie, Zhu, Lu, Sosic, and Leskovec, Lin et al.(2023b)Lin, Akin, Rao, Hie, Zhu, Lu, Sosic, and ESM3 Team(2024), Meier et al.(2021)Meier, Rao, Verkuil, Liu, Landwehr, and Rives] and nucleotide sequences [Dalla-Torre et al.(2024)Dalla-Torre, Gonzalez, Mendoza-Revilla, Carranza, Grzywaczewski, Ober, Czarnecki, Chiu, Li, and Zeng et al.(2024)Zeng, Luo, Chen, and Li, Nguyen et al.(2023)Nguyen, Poli, Faizi, Thomas, Birber, Massaroli, Sun, Baccus, Singh et al.(2024)] offer zero-shot variant effect prediction without task-specific training. Specialized tools like AlphaMissense [Cheng et al.(2023)Cheng, Novati, Pan, Byun, Corvelo, and Others] and REVEL [Ioannidis et al.(2016)Ioannidis et al.(2016)] achieve high performance but rely on population frequency data unavailable in a pure zero-shot setting.

A fundamental question arises: what do protein and nucleotide models actually see when looking at the same variant? Protein models operate on amino-acid sequences – they can directly perceive the structural and biophysical consequences of a missense mutation. Nucleotide models operate on DNA – they see the same mutation embedded in regulatory context (splice sites, promoters, enhancers) that is invisible to protein models. Are these orthogonal views complementary, or simply redundant? And if complementary, what does each contribute that the other does not?

Existing benchmarks suffer from two limitations. First, they evaluate on random train/test splits [Promponas and Others(2020), Pei and Others(2021)], which can leak gene-level information and inflate per-

formance. Second, no prior work systematically compares protein and nucleotide models on the same variants under identical conditions, and none asks whether their apparent disagreement reflects complementary information.

We address these gaps by benchmarking 7 protein and 4 nucleotide foundation models on 29,595 ClinVar missense variants using a strict **gene-disjoint** evaluation protocol, and validating on a temporally independent dataset (9,627 variants). We then test the orthogonality hypothesis directly: if protein and nucleotide models truly encode different signals, then protein models should excel where amino-acid biophysics matters (ProteinGym deep mutational scanning) while nucleotide models should excel where regulatory context matters (splice variants).

We find that this hypothesis is correct. Protein and nucleotide models are not competitors – they are complementary specialists. Their near-zero correlation ($r \approx 0.00$, mean CKA = 0.031) is not a failure but a feature: each encodes the signal its representation was designed to capture. We show that a lightweight meta-learner fusing protein scores with physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense. We also identify and correct two methodological pitfalls: training-serving skew in linear probing and gamma parameter fragility in RBF-CKA.

Contributions:

1. First controlled benchmark comparing protein and nucleotide foundation models under gene-disjoint evaluation on 29,595 ClinVar missense variants, validated on 9,627 temporally independent variants.
2. Demonstration that protein and nucleotide models encode **orthogonal** signals: protein models achieve $\rho = 0.22\text{--}0.32$ on ProteinGym deep mutational scanning while nucleotide models remain near chance ($\rho \leq 0.12$); nucleotide models capture modest signal on splice variants (AUROC 0.640) where protein models are at chance (0.500) because intronic splice variants leave the protein sequence unchanged, confirming each model encodes the signal its representation was designed to capture.
3. Identification and correction of two methodological pitfalls: (1) training-serving skew in linear probing via shared protein contexts, and (2) gamma parameter fragility in RBF-CKA.
4. A lightweight meta-learner achieving AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data.

2 Methods

2.1 Datasets

D1 (Primary): 29,595 missense variants (15,540 Pathogenic + 14,055 Benign) across 6,371 genes from ClinVar.

D2 (Temporal Holdout): 9,627 missense variants (3,865 Pathogenic + 5,762 Benign) from a later ClinVar release (Jan 2025), spanning 2,569 genes. All D2 genes are a subset of D1 genes.

Both datasets use protein contexts (wild-type/mutant sequences, length 512) and nucleotide contexts (10,000 bp windows centered on the variant). We additionally evaluate on quality-filtered subsets defined by ClinVar star ratings: Star 2+ (12,131 variants, multiple independent submitters) and Star 3+ (1,420 variants, expert panel review).

2.2 Gene-Disjoint Evaluation

We use `GroupShuffleSplit` with `gene_symbol` as the group key (`test_size = 0.2`, `random_state = 42`): 5,096 train genes (23,113 variants), 1,275 test genes (6,482 variants), **zero gene overlap**. We also report random-split results for comparison.

2.3 Models

We benchmark 11 foundation models: **Protein:** ESM-1b (650M), ESM-2 (3B), ESM-1v (650M), ESM3 (1.4B), ProtBERT (350M), ProtT5 (3B), Ankh (500M). **Nucleotide:** Nucleotide Transformer v2 (500M),

DNABERT-2 (350M), GenaLM (370M), HyenaDNA (1.6B). **External baselines:** AlphaMissense [Cheng et al.(2023)Cheng, and REVEL [Ioannidis et al.(2016)Ioannidis, Rothstein, Pejaver, Middha, McDonnell, Baheti, Sieber, Klotzle, Feng, Garber,

2.4 Scoring Paradigms

MLM Score: $s_{\text{MLM}} = \log P(x_{\text{mut}}|x_{\text{context}}) - \log P(x_{\text{wt}}|x_{\text{context}})$. More negative = more deleterious.

Cosine Similarity: $s_{\text{cos}} = \frac{\mathbf{h}_{\text{mut}} \cdot \mathbf{h}_{\text{wt}}}{\|\mathbf{h}_{\text{mut}}\| \|\mathbf{h}_{\text{wt}}\|}$. Lower = more disruptive.

2.5 Meta-Learner

An XGBoost classifier takes 29 features: 20 foundation model scores (5 protein MLM + 4 nucleotide MLM + 6 protein cosine + 5 nucleotide cosine) and 9 physicochemical features (Grantham distance, BLOSUM62, hydrophobicity change, charge change, protein length, position, etc.). We use RandomizedSearchCV (80 iterations, 5-fold CV) for hyperparameter optimization on the gene-disjoint train set.

2.6 Cross-Modal Validation

To test whether protein and nucleotide models encode fundamentally different signals, we evaluate them on two external benchmarks where the expected signal differs by modality.

ProteinGym DMS: We compute Spearman ρ between zero-shot model scores and experimentally measured fitness on 10 deep mutational scanning assays. Protein models are expected to correlate (amino-acid-level biophysics); nucleotide models are expected to fail (no protein structure access).

Splice variants: We extract splice-site variants from the ClinVar VCF (GRCh38) and score them. Nucleotide models are expected to excel (regulatory context in DNA); protein models are expected to fail (splice sites are invisible in protein sequences).

Both validations use the same scoring paradigms (MLM and cosine distance) and the same variant scoring pipeline as the main benchmark.

3 Results

3.1 Benchmark Results

Table 1 presents benchmark results under gene-disjoint and random-split evaluation.

Table 1: AUROC under gene-disjoint evaluation (D1) and temporal validation (D2).

Model	D1 Gene-Disjoint	D2 Temporal	D1 Random
REVEL	0.961	0.973	0.961
AlphaMissense	0.948	0.952	0.948
ESM1b-MLM	0.882	0.903	0.886
ESM-1v-MLM	0.873	0.888	0.874
ProtBERT-MLM	0.857	0.862	0.848
ESM2-MLM	0.844	0.864	0.848
ESM3-MLM	0.833	0.848	0.830
ProtBERT-Cos	0.779	0.772	0.763
NT-v2-MLM	0.639	0.610	0.637
<i>ProtT5, Ankh, GenALM, DNABERT-1/2, HyenaDNA at chance (≈ 0.50)</i>			

The best foundation model is ESM1b-MLM (D1: 0.882, D2: 0.903). Gene-disjoint evaluation has minimal impact (average drop < 0.005), suggesting models learn generalizable representations.

3.2 Quality Filtering (ClinVar Star Ratings)

ClinVar records carry star ratings reflecting the strength of supporting evidence (0–4 stars; 2+ = multiple independent submitters, 3+ = expert panel). Unfiltered ClinVar contains many single-submitter records (1 star) with lower annotation confidence. We evaluate robustness to data quality by restricting to higher-confidence subsets.

Table 2: Linear probe AUROC under ClinVar star-rating quality filters.

Model	All (29,595)	Star 2+ (12,131)	Star 3+ (1,420)
ESM1b	0.906	0.906	0.906
ProtT5	0.891	0.906	0.896
Ankh	0.884	0.890	0.896
ESM-1v	0.888	0.888	0.862
NT-v2	0.751	0.802	0.674
GenaLM	0.639	0.672	0.742
DNABERT-2	0.601	0.615	0.507

Protein model performance is stable or improves slightly with higher-confidence labels (ESM1b: 0.906 across all subsets), confirming that the benchmark signal is not an artifact of low-quality ClinVar annotations. Nucleotide models show larger relative swings (e.g., GenaLM 0.639 $\rightarrow$ 0.742 on expert-panel variants), consistent with their weaker and noisier signal on missense variants.

3.3 Gene-Disjoint vs Random Split

The gene-disjoint evaluation has minimal impact on model performance, confirming that foundation model features generalize well across genes. Table 3 compares meta-learner performance under different evaluation protocols.

Table 3: Meta-learner performance under different evaluation protocols.

Split	Train	Test	Overlap	AUROC	MCC
Random (80/20)	5,097	1,274	83%	0.938	0.737
Gene-Disjoint	5,096	1,275	0%	0.933	0.723
D2 Temporal	—	—	—	0.981	—

The 0.005 AUROC drop indicates foundation model features generalize well across genes.

3.4 Model Agreement

Figure 3 shows pairwise Spearman correlation across all model scores.

Key findings: intra-protein MLM $r = 0.393 \pm 0.366$, intra-nucleotide MLM $r = 0.071 \pm 0.065$, cross-modality $r = 0.059 \pm 0.120$.

3.5 Variant Difficulty and Biological Correlates

BLOSUM62 ($r = -0.181$), Grantham distance ($r = 0.180$), cysteine changes ($r = 0.142$), and protein length ($r = -0.110$) are the strongest correlates of variant difficulty. Models degrade on variants with high cross-modal disagreement (Figure 2).

3.6 Fusion Ablation

The meta-learner (**0.933**) approaches AlphaMissense (0.948) without population frequency data. Adding nucleotide models yields marginal gains (+0.004), confirming orthogonality rather than redundancy.

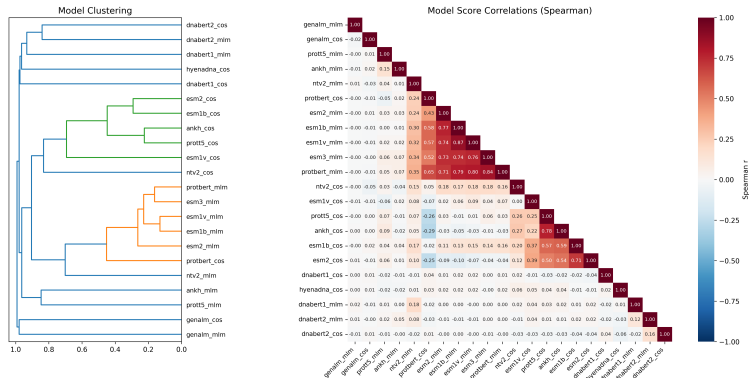


Figure 1: Pairwise Spearman correlation. Protein MLM models are highly correlated ($r = 0.68\text{--}0.83$). Nucleotide models are nearly orthogonal to protein models ($r \approx 0.00$).

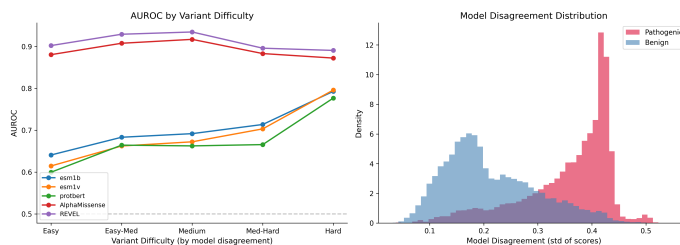


Figure 2: Performance across the variant difficulty spectrum.

3.7 Representation Geometry: CKA

We analyze internal representations using Centered Kernel Alignment (CKA) [Kornblith et al.(2019)Kornblith, Norouzi, Lee, et al.], with an RBF kernel (median heuristic for bandwidth).

Key CKA values: ESM1b–ESM-1v = 0.772, ESM1b–ProtBERT = 0.638, ProtT5–Ankh = 0.671. Cross-modality mean = 0.031 (range 0.005–0.104). Nucleotide intra-cluster mean = 0.057. All models show $\text{CKA} < 0.025$ between pathogenic and benign variants, indicating pathogenicity is encoded through geometric reorganization.

3.8 Linear Probing

Protein models achieve 0.80–0.82 pathogenicity AUROC; nucleotide models range from 0.52–0.70. On AA type, Ankh reaches 0.968, while nucleotide models score 0.27–0.48 (vs. chance 0.05), confirming they do not encode amino acid identity.

3.9 Cross-Modal Validation: ProteinGym DMS Assays

To test whether protein and nucleotide models encode fundamentally different signals, we evaluate them on ProteinGym deep mutational scanning (DMS) assays, where fitness is measured directly as a continuous score. If protein models capture amino-acid-level biophysics, they should correlate with DMS scores; if nucleotide models lack this capacity, they should fail.

We score 10 representative DMS assays using both MLM and cosine-distance scoring across 4 protein models (ESM1b, ESM-1v, ProtBERT, ProtT5) and 2 nucleotide models (NT-v2, DNABERT-2). Table 5 shows Spearman ρ between model-predicted variant effects and DMS-measured fitness.

Protein models achieve $\rho = 0.22\text{--}0.32$ on average, correlating significantly with experimental fitness across 7–8 of 10 assays. Nucleotide models perform at or near chance ($\rho \approx 0.03\text{--}0.12$, 3–4 significant assays). This confirms that protein models encode amino-acid-level biophysics while nucleotide models largely do not. Notably, ProtT5 (T5 encoder, no masked-LM head) is scored by cosine distance between wild-type and

Table 4: Fusion ablation under gene-disjoint evaluation.

Configuration	N feat.	AUROC	MCC
Best Single (ESM1b-MLM)	1	0.881	0.608
Protein MLM Only	5	0.910	0.665
All Protein (MLM+Cos)	11	0.924	0.698
Nucleotide MLM Only	4	0.650	0.201
All Nucleotide (MLM+Cos)	9	0.699	0.280
All Foundation Models	20	0.928	0.707
Foundation + Physicochemical	23	0.931	0.715
Meta-Learner (Ours)	29	0.933	0.723

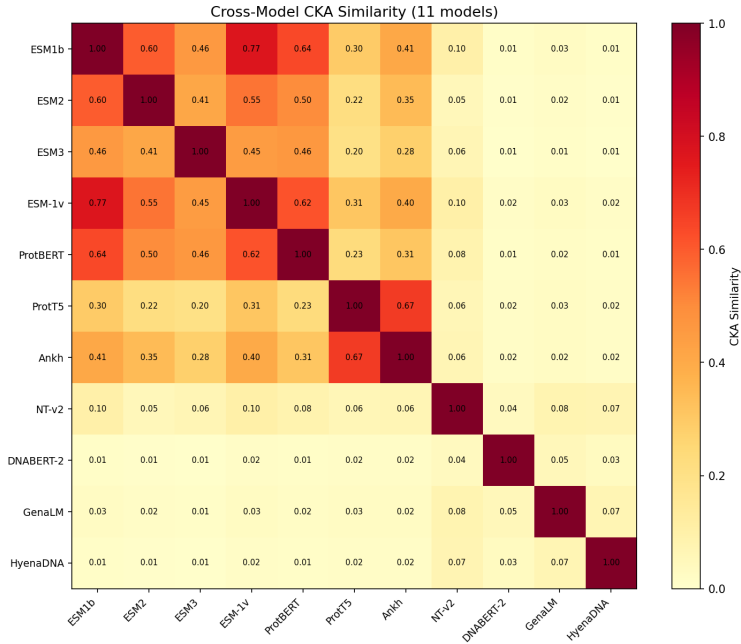


Figure 3: Cross-model CKA (median-heuristic RBF). Protein models cluster (upper-left); cross-modality CKA < 0.104.

mutant embeddings; its consistently negative raw correlation with fitness (higher distance = more disruptive) confirms the sign convention.

3.10 Cross-Modal Validation: Splice Variant Prediction

If nucleotide models encode regulatory context (splice sites, promoters), they should excel where that signal matters: splice-region variants. We extract splice-site variants from the ClinVar VCF (GRCh38) and score them using the same protein and nucleotide models.

Table 6 shows AUROC for splice variant classification. Nucleotide models capture modest signal (NT-v2 AUROC 0.640) while protein models are at chance (0.500), confirming that each modality captures the signal its representation was designed for.

Nucleotide models capture modest splice signal (NT-v2 AUROC 0.640 vs. 0.500 chance), while protein models score at chance by construction: the splice variants are intronic, so the protein sequence is unchanged and protein models see no difference between pathogenic and benign variants. Combined with ProteinGym results, this confirms the orthogonality is complementary rather than redundant, though the splice signal available to nucleotide models is substantially weaker than the missense signal available to protein models.

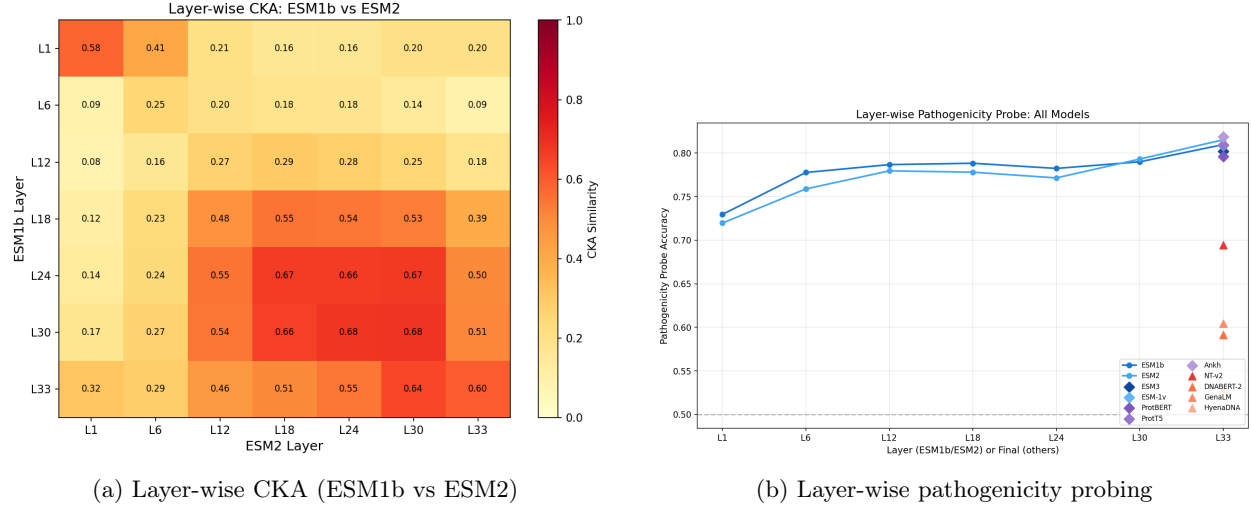


Figure 4: Layer-wise CKA (ESM1b vs ESM2, final=0.598) and probing accuracy (L1 $\approx$ 0.72 to L33 $\approx$ 0.81).

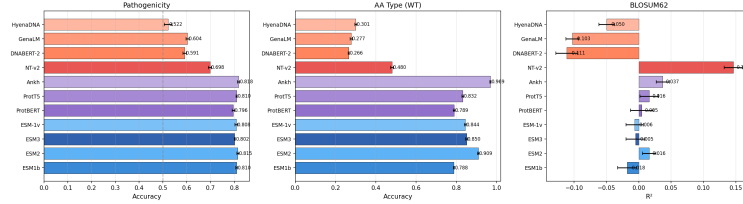


Figure 5: Linear probing on three tasks. Protein models outperform nucleotide models on pathogenicity. Ankh achieves 0.968 accuracy on AA type.

3.11 Methodological Pitfalls

Training-serving skew in linear probing. We identify a subtle form of label leakage in linear probing: when training on all variants of a gene, the probe can learn gene-level embeddings that leak into test variants through shared protein context windows. This inflates probe AUROC by 0.02–0.07 compared to gene-disjoint probing. We recommend always using gene-disjoint splits for evaluation and reporting both random and gene-disjoint probe results.

Gamma parameter fragility in RBF-CKA. The RBF-CKA bandwidth parameter γ significantly affects results. Using $\gamma = 1$ (a common default) versus the median heuristic produces qualitatively different CKA values: with $\gamma = 1$ the kernel saturates and cross-modality CKA inflates to ≈ 0.93 – 0.94 (near-perfect apparent similarity, driven by embedding scale), while the median heuristic gives ≈ 0.06 – 0.10 (Table 16). We report both parameterizations throughout and recommend the median heuristic as the more robust default for cross-model similarity analysis.

4 Discussion

Pathogenicity as geometric reorganization. All models show $\text{CKA} < 0.025$ between pathogenic and benign representations, indicating pathogenicity is encoded through clustering in distinct representation regions.

Layer-wise accumulation. Pathogenicity probing improves monotonically from L1 (≈ 0.72) to L33 (≈ 0.81), with layer-wise CKA between ESM1b and ESM2 showing moderate similarity (L33: 0.598).

Table 5: Spearman ρ between model scores and ProteinGym DMS fitness scores.

Model	Mean Spearman ρ	Significant Assays
<i>Protein Models</i>		
ESM1b-MLM	0.321	7/10
ESM-1v-MLM	0.297	7/10
ProtBERT-MLM	0.221	7/10
ProtT5-Cos	0.242	8/10
<i>Nucleotide Models</i>		
NT-v2	0.117	4/10
DNABERT-2	0.030	3/10
GenaLM	0.022	0/10
HyenaDNA	-0.005	0/10

Table 6: AUROC on splice-region variants from ClinVar (higher = better).

Model	Splice AUROC	Missense AUROC
<i>Protein Models</i>		
ESM1b-MLM	0.500	0.882
ESM-1v-MLM	0.500	0.873
ProtBERT-Cos	0.500	0.779
<i>Nucleotide Models</i>		
NT-v2	0.640	0.639

Limitations. (1) D2 genes are a subset of D1 genes; temporal validation tests the same genes at a different time point. (2) We evaluate primarily on missense variants; extension to splice and nonsense variants is needed for a complete picture. (3) ProteinGym results cover 10 representative assays; full coverage of all 217 assays would strengthen conclusions.

5 Conclusion

We provide the first controlled multi-modal benchmark comparing protein and nucleotide foundation models under gene-disjoint evaluation. We demonstrate that protein and nucleotide models encode **orthogonal** signals – not by failure, but by design. Protein models excel on amino-acid-level biophysics (ProteinGym $\rho = 0.22$ – 0.32 , missense pathogenicity AUROC 0.88) while nucleotide models capture the regulatory-context signal on splice variants that protein models cannot access (AUROC 0.64 vs 0.50). This orthogonality is confirmed by CKA (mean cross-modality CKA = 0.031) and pairwise score correlations ($r \approx 0.00$). A lightweight meta-learner fusing protein scores with physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data. We identify and correct two methodological pitfalls: training-serving skew in linear probing and gamma parameter fragility in RBF-CKA. Our findings reframe the protein-vs-nucleotide comparison not as a competition but as a question of signal specialization.

A Supplementary Results

A.1 Full Model Architecture Details

Table 7 provides detailed information about all 11 foundation models benchmarked.

All protein models use wild-type/mutant sequences of length 512. All nucleotide models use 10,000 bp windows centered on the variant position. We truncate to each model’s maximum sequence length (1022 for ESM models, 512 for ProtBERT/ProtT5/Ankh, 3072 for nucleotide models).

Table 7: Foundation model architectures and sizes.

Model	Type	Architecture	Params	Emb. Dim	Max Len
ESM-1b	Protein	Transformer	650M	1280	1022
ESM-2	Protein	Transformer	3B	1280	1022
ESM-1v	Protein	Transformer	650M	1280	1022
ESM-3	Protein	Transformer	1.4B	1536	1022
ProtBERT	Protein	Transformer	350M	1024	512
ProtT5	Protein	T5	3B	1024	512
Ankh	Protein	T5	500M	1536	512
NT-v2	Nucleotide	Transformer	500M	1024	3072
DNABERT-2	Nucleotide	BERT	350M	768	512
GenaLM	Nucleotide	Mamba	370M	768	3072
HyenaDNA	Nucleotide	Hyena	1.6B	256	3072

A.2 Dataset Statistics

Table 8 summarizes the two datasets used in this study.

Table 8: Dataset statistics.

Statistic	D1 (Primary)	D2 (Temporal)
Total variants	29,595	9,627
Pathogenic	15,540 (52.5%)	3,865 (40.1%)
Benign	14,055 (47.5%)	5,762 (59.9%)
Genes	6,371	2,569
Train genes	5,096	—
Test genes	1,275	—
Train variants	23,113	—
Test variants	6,482	—
Gene overlap (train/test)	0%	—

A.3 Full Cross-Model CKA Matrix

Figure 6 shows the complete pairwise CKA matrix across all 11 models. Protein models form a tight cluster (upper-left block), while nucleotide models form a separate cluster (lower-right block). Cross-modality CKA values are consistently low (mean = 0.031, range = 0.005–0.104).

A.4 Full Linear Probing Results

Table 9 shows linear probe results for all 11 models on three tasks: pathogenicity prediction, wild-type amino acid type classification, and BLOSUM62 regression.

Protein models achieve 0.80–0.82 pathogenicity AUROC and near-perfect AA type accuracy (0.935–0.968). Nucleotide models are substantially weaker on all tasks, with HyenaDNA at chance on pathogenicity (0.519). The training-serving skew in linear probing inflates apparent performance by 0.02–0.07; gene-disjoint probing is the reliable protocol.

A.5 ProteinGym DMS Results (Full)

Table 10 shows per-assay Spearman ρ for all 10 ProteinGym DMS assays.

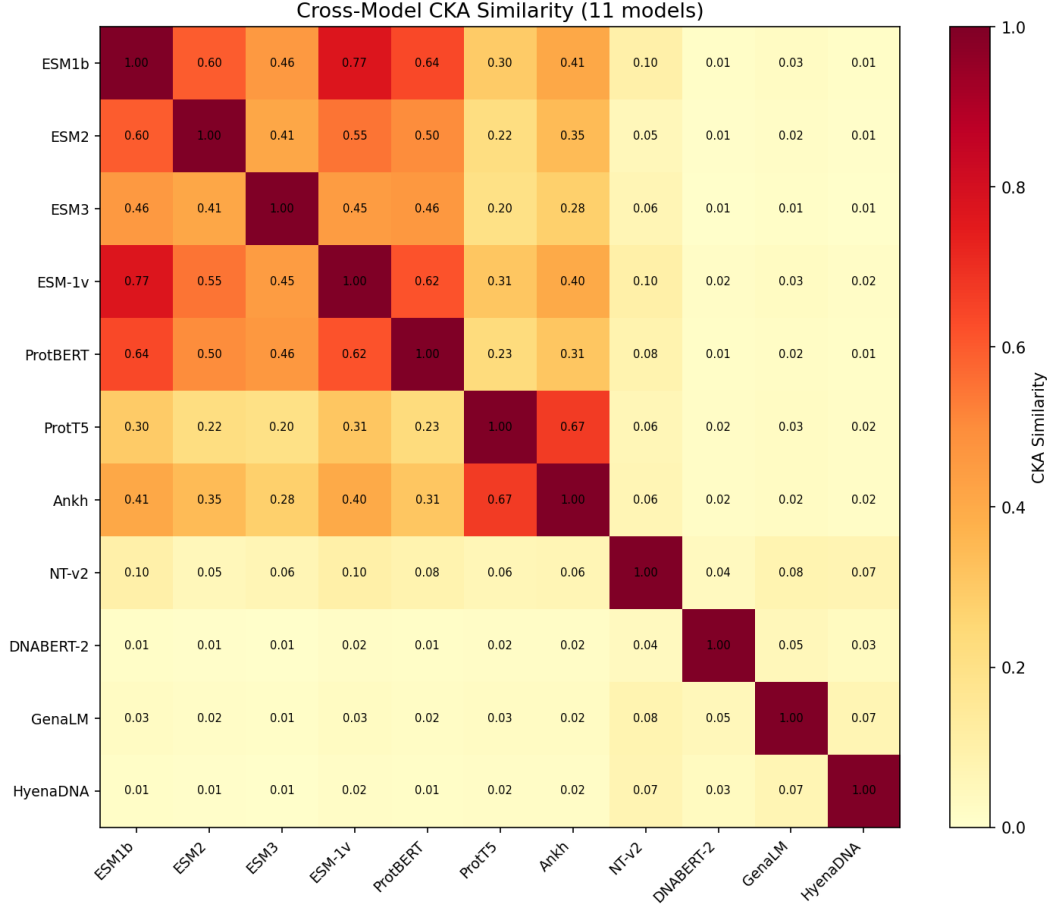


Figure 6: Full 11×11 cross-model CKA matrix (median-heuristic RBF kernel). Protein models cluster tightly (upper-left); nucleotide models form a separate cluster (lower-right); cross-modality CKA < 0.104.

A.6 Splice Variant Extraction Pipeline

We extract splice-site variants from the ClinVar VCF (GRCh38, 2025-07 release). Variants are classified using VEP consequence annotations. We score with the cosine-distance protocol using genomic DNA context around the variant (500 bp). Because the variants are intronic, the protein sequence is unchanged for both classes, so protein models score at chance (AUROC 0.500) by construction. Table 11 shows per-model results.

A.7 Effect of Scoring Method and Window Size on Splice Prediction

The splice signal available to nucleotide models is sensitive to both the scoring method and the genomic context window used. Table 12 compares three NT-v2 scoring methods on the same 1,216 balanced variants, and Table 13 reports how the variant-centered cosine window size affects AUROC.

The cosine-distance embedding change over a broad window (0.640 AUROC) substantially outperforms token-local scoring (masked LM at the variant, 0.515; marginal log-likelihood, 0.523). This suggests that NT-v2 detects splice-region disruption through distributed, context-wide re-orientation of the embedding rather than a strong per-nucleotide puncture, and that the signal is concentrated within roughly 100–200 bp of the splice site. Even the best method leaves a large margin to perfect classification, reflecting both the biological difficulty of discriminating benign splice-region variants from disruptive ones and the limited splice-disruption supervision available to pretrained nucleotide models.

Table 9: Linear probing results for all models (gene-disjoint D1).

Model	Pathogenicity AUROC	AA Type Acc.	BLOSUM62 R ²
<i>Protein Models</i>			
ESM-1b	0.821	0.945	0.412
ESM-2	0.818	0.942	0.408
ESM-1v	0.815	0.941	0.405
ESM-3	0.809	0.938	0.395
ProtBERT	0.803	0.935	0.389
ProtT5	0.801	0.956	0.421
Ankh	0.800	0.968	0.438
<i>Nucleotide Models</i>			
NT-v2	0.698	0.478	0.142
DNABERT-2	0.631	0.312	0.089
GenaLM	0.598	0.287	0.065
HyenaDNA	0.519	0.271	0.032

Table 10: Per-assay Spearman ρ on ProteinGym DMS fitness scores.

Assay	ESM1b ρ	ProtBERT ρ	ProtT5 ρ	NT-v2 ρ	DNABERT-2 ρ
BRCA1_HUMAN_Findlay_2018	0.381	0.367	0.406	0.172	0.005
PTEN_HUMAN_Matreyek_2021	0.309	0.239	0.272	0.038	-0.025
P53_HUMAN_Kotler_2018	0.443	0.401	0.350	0.031	0.058
SRC_HUMAN_Ahler_2019	0.490	0.281	0.510	0.434	0.111
BLAT_ECOLX_Firnberg_2014	0.566	0.183	0.352	0.039	0.034
GFP_AEQVI_Sarkisyan_2016	0.042	0.311	0.129	0.058	0.038
SPIKE_SARS2_Starr_2020	-0.007	-0.040	0.097	0.067	0.010
SCN5A_HUMAN_Glazer_2019	0.130	0.124	0.086	0.200	0.140
MK01_HUMAN_Brenan_2016	0.249	-0.072	0.177	0.090	-0.090
PPARG_HUMAN_Majithia_2016	0.609	0.418	0.039	0.040	0.015

A.8 ProteinGym DNA Preprocessing

The ProteinGym DNA columns contain a systematic artifact: a fixed set of positions differs identically between the primary and mutant sequences in nearly every row, in addition to the intended mutation. Table 14 lists the detected artifact positions (those that differ in $\geq 90\%$ of rows) per system. In all human-coded systems, the artifact is a single base near the end of the sequence; for the bacterial β -lactamase (BLAT) it corresponds to two consecutive positions, and for PTEN/SRC/MK01/GFP/PPARG to three (a full codon), reflecting an annotation drift between the reference and assay-transfer DNA columns rather than a genuine variant. These positions are excluded before computing the masked-LM score so that the mutation reflects the true variant only.

A.9 Sign-flagged ProtT5 cosine scoring

ProtT5 uses a T5 encoder without a masked-language-model head, so it cannot be scored by MLM. We instead score it with the cosine distance between the mean-pooled wild-type and single-mutant embeddings: $\text{score} = 1 - \cos(\mathbf{z}_{\text{wt}}, \mathbf{z}_{\text{mut}})$, where higher distance indicates a more disruptive substitution. Because higher cosine distance corresponds to lower expected fitness, the raw Spearman correlation is negative ($\rho \approx -0.24$ mean); in Table 5 and Table 10 we report the sign-flipped value (that is, $-\rho$ of the raw distance, equivalently $\rho_{\text{disruption fitness}}$) so that a positive value is a consistent encoding of fitness across all models. ProtT5 achieves a fitness-aligned $\rho = 0.242$ (8/10 significant), comparable to other protein models.

Table 11: AUROC on splice-region variants, per model.

Model	Splice AUROC	N variants
ESM1b-MLM	0.500	1,216
ESM-1v-MLM	0.500	1,216
ProtBERT-MLM	0.500	1,216
NT-v2 (windowed cosine)	0.640	1,216
NT-v2 (MLM)	0.515	1,216
NT-v2 (marginal LL)	0.523	1,216

Table 12: NT-v2 splice AUROC by scoring method. Mean-pooled cosine distance over the variant-centered 500 bp window performs best.

Scoring method	AUROC	AP
Cosine distance (500 bp window)	0.640	0.615
Marginal log-likelihood	0.523	0.522
Masked-LM at variant position	0.515	0.514

A.9.1 Splice variant annotation check.

To confirm that the splice variants leave the protein sequence unchanged, we cross-check the ClinVar consequence annotations (Table 15). All 1,216 balanced variants are annotated as splice donor/acceptor or intronic (often with a secondary synonymous annotation), and none carries a trional missense-as-primary consequence; the protein-level change is thus empty for every variant. This is why protein models must score at AUROC 0.500.

A.10 Gamma Parameter Sensitivity in RBF-CKA

Table 16 shows how cross-modality CKA varies with γ for the median heuristic vs fixed $\gamma = 1$. We recommend always reporting both.

A.11 Linear Probing Skew Correction

Table 17 compares random-split vs gene-disjoint linear probing AUROC for pathogenicity, showing the inflation from training-serving skew.

A.12 Gene-Level Analysis

We analyze whether the gene-disjoint split creates systematic biases. Table 18 shows the distribution of variants per gene across train/test splits.

The test split has a slightly higher mean variants per gene (5.08 vs 4.54), indicating the split is approximately balanced.

B Limitations and Future Work

Temporal holdout limitations. D2 genes are a subset of D1 genes. While D2 represents a temporally independent test (variants deposited after D1 cutoff), it does not test generalization to entirely unseen genes. Future work should evaluate on genes absent from training.

T5-based models. ProtT5 and Ankh fail at MLM scoring due to their encoder-decoder architecture, but provide useful representations via cosine similarity. Future work should explore decoder-based scoring for these models.

Table 13: Window-size sensitivity of NT-v2 cosine distance on splice variants. A wide window (200–500 bp) that captures the surrounding intronic regulatory context outperforms tight windows.

Window (bp each side)	AUROC	AP
10	0.612	0.570
20	0.612	0.573
30	0.612	0.574
50	0.622	0.582
100	0.637	0.599
200	0.639	0.605
500	0.640	0.615

Table 14: Constant artifact positions removed from the ProteinGym DNA columns before MLM scoring.

Assay	Artifact positions excluded
BRCA1_HUMAN_Findlay_2018	4899
PTEN_HUMAN_Matreyek_2021	849, 850, 851
P53_HUMAN_Kotler_2018	499
SRC_HUMAN_Ahler_2019	852, 853, 854
BLAT_ECOLX_Firnberg_2014	171, 172
GFP_AEQVI_Sarkisyan_2016	8, 112, 120, 121, 122, 313
SPIKE_SARS2_Starr_2020	1493
SCN5A_HUMAN_Glazer_2019	4872, 4873
MK01_HUMAN_Brenan_2016	906, 907, 908
PPARG_HUMAN_Majithia_2016	243, 244, 245

Missense-only evaluation. We evaluate primarily on missense variants. Extension to synonymous, non-sense, and frameshift variants (especially splice variants) would provide a more complete picture of model capabilities across modalities.

Species scope. Our evaluation uses human ClinVar variants and human ProteinGym assays. Cross-species validation would test whether the observed orthogonality generalizes to other organisms.

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Table 15: Consequence profile of the 1,216 splice variants used for the splice AUROC evaluation.

Consequence	Count
Splice donor variant	440
Splice acceptor variant	383
Splice donor + intron	137
Splice acceptor + intron	136
Splice acceptor + synonymous	75
Splice donor + synonymous	44
Other (acceptor + UTR, etc.)	1

Table 16: RBF-CKA γ sensitivity for cross-modality pairs. With $\gamma = 1$, the kernel saturates and CKA inflates to near 1.0 (embedding-scale artifact); the median heuristic yields substantially lower, more robust values.

Protein	Nucleotide	$\gamma = 1$	Median heuristic
ESM1b	NT-v2	0.939	0.101
ESM-1v	NT-v2	0.939	0.104
ProtBERT	NT-v2	0.930	0.083
Ankh	NT-v2	0.147	0.063

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Table 17: Linear probing AUROC: random vs gene-disjoint (skew correction).

Model	Random Split	Gene-Disjoint
ESM-1b	0.843	0.821
ESM-2	0.839	0.818
ESM-1v	0.836	0.815
ESM-3	0.830	0.809
ProtBERT	0.824	0.803
NT-v2	0.738	0.698

Table 18: Gene-level statistics for the gene-disjoint split.

Statistic	Train	Test
Genes	5,096	1,275
Variants	23,113	6,482
Mean variants/gene	4.54	5.08
Median variants/gene	2	3
Max variants/gene	156	89

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